Earners AI and ML by Aditya Jain sir - Lecture no. 20

OOP Recap

```
class Student:

  def __init__(self, name, age):
    self.name = name
    self.age = age
```

```
s1 = Student("Raj", 22)
```

```
print(s1.name)
```

```
Raj
```

```
print(s1.age)
```

```
22
```

```
s2 = Student("Rahul", 25)
```

```
print(s2.name)
```

```
Rahul
```

```
print(s2.age)
```

```
25
```

```
class Student:

  def __init__(self, name, age):
    self.name = name
    self.age = age

  def show(self):
    print("name:", self.name, "age:", self.age)
```

```
s1 = Student("Raj", 22)
```

```
s2 = Student("Rahul", 25)
```

```
s1.show()
```

```
name: Raj age: 22
```

```
s2.show()
```

```
name: Rahul age: 25
```

HW

```
class Mobile:

    def __init__(self, brand):
        self.brand = brand

m = Mobile("Samsung")

print(hasattr(m, "brand"))
```

```
True
```

```
print(hasattr(m, "name"))
```

```
False
```

```
# What is the output?

class Student:

    def show(self):
        print("Python")

s = Student()

print(type(s))

A.

<class '__main__.Student'>

B.

Student

C.

Python

D. Error
```

```
class Student:

    def show(self):
        print("Python")

s = Student()

print(type(s))
```

```
<class '__main__.Student'>
```

```
class Book:

    def __init__(self, title):
        self.title = title

b = Book("Python Basics")
```

```
# What is the output?

class Book:

    def __init__(self, title):
        self.title = title

b = Book("Python Basics")

print(len(b.title))
```

```
13
```

```
b.title
```

```
'Python Basics'
```

```
# What is the output?

class Rectangle:

    def __init__(self, length, width):
        self.length = length
        self.width = width
```

```
    def area(self):
        return self.length * self.width

r = Rectangle(5, 4)

print(r.area())
```

```
20
```

What is the output?

```
class Circle:

    def __init__(self, radius):
        self.radius = radius

c1 = Circle(5)
c2 = Circle(10)

print(c1.radius + c2.radius)
```

```
class Circle:

    def __init__(self, radius):
        self.radius = radius

c1 = Circle(5)
c2 = Circle(10)

print(c1.radius + c2.radius)
```

```
15
```

What is the output?

```
class Laptop:

    def __init__(self, brand):
        self.brand = brand

    def display(self):
        print(self.brand)

l = Laptop("Dell")

l.display()
l.display()
```

```
class Laptop:

    def __init__(self, brand):
        self.brand = brand

    def display(self):
        print(self.brand)

l = Laptop("Dell")

l.display()
l.display()
```

```
Dell
Dell
```

```
class Bank:

    def __init__(self, balance):
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount

account = Bank(1000)
```

```
account.deposit(500)

print(account.balance)
```

```
1500
```

```
class Student:

    def __init__(self, name, marks):
        self.name = name
        self.marks = marks

    def result(self):
        if self.marks >= 40:
            print("Pass")
        else:
            print("Fail")

s = Student("Rahul", 35)

s.result()
```

```
Fail
```

```
class Student:

    def __init__(self, name):
        self.name = name

    def change_name(self, new_name):
        self.name = new_name

s = Student("Rahul")


s.change_name("Amit")

print(s.name)
```

```
Amit
```

```
class Student:

    def __init__(self, name):
        self.name = name

    def change_name(self, new_name):
        self.name = new_name

s = Student("Rahul")
print(s.name)

s.change_name("Amit")
print(s.name)
```

```
Rahul
Amit
```

```
class Counter:

    def __init__(self):
        self.count = 0

    def increase(self):
      print(self.count)
      self.count += 1

c = Counter()

for i in range(5):
    c.increase()

print(c.count)
```

```
0
1
2
3
4
5
```

class variable and instance variable

```
class Student:

  college = "IIT Bombay"

  def __init__(self, name):
    self.name = name
```

```
s1 = Student("Raj")
s2 = Student("Rahul")
s3 = Student("Nimisha")
```

```
print(s1.name, s1.college)
```

```
Raj IIT Bombay
```

```
print(s2.name, s2.college)
```

```
Rahul IIT Bombay
```

```
print(s3.name, s3.college)
```

```
Nimisha IIT Bombay
```

```
class Student:

  college = "IIT Bombay"

  def __init__(self, name):
    self.name = name
```

```
s1 = Student("Raj")
s2 = Student("Rahul")
s3 = Student("Nimisha")
```

```
print(s1.name, s1.college)
```

```
Raj IIT Bombay
```

```
s1.college = "IIT Madras"
```

```
print(s1.name, s1.college)
```

```
Raj IIT Madras
```

```
print(s2.name, s2.college)
```

```
Rahul IIT Bombay
```

```
print(s3.name, s3.college)
```

```
Nimisha IIT Bombay
```

Inheritance

```
parent class --> the class we are inheriting from --> base class

child class --> the class that inherits from another class --> derived class
```

```
class Person:
  def __init__(self, first_name, last_name):
    self.first_name = first_name
    self.last_name = last_name

  def show_name(self):
    print(self.first_name, self.last_name)
```

```
p1 = Person("Raj", "Sharma")
p1.show_name()
```

```
Raj Sharma
```

```
class Student(Person):
  pass
```

```
s1 = Student("Raj", "Sharma")
```

```
s1.show_name()
```

```
Raj Sharma
```

```
# Create a parent class Vehicle with an __init__ method that takes brand as input.

# Add a method display_brand() that prints the vehicle's brand.

# Create a child class Car that inherits from Vehicle.
```

```
class Vehicle:
  def __init__(self, brand):
    self.brand = brand

  def display_brand(self):
    print("Brand of the car is", self.brand)
```

```
class Car(Vehicle):
  pass
```

```
c1 = Car("Suzuki")
```

```
c1.display_brand()
```

```
Brand of the car is Suzuki
```

```
class Vehicle:
  def __init__(self, brand):
    self.brand = brand

  def display_brand(self):
    print("Brand of the car is", self.brand)
```

```
class Car(Vehicle):
  car_engine = "kia_1"
```

```
c2 = Car("Kia")
```

```
c2.display_brand()
```

```
Brand of the car is Kia
```

```
c2.car_engine
```

```
'kia_1'
```

Polymorphism

```
s = "Aditya"
```

```
len(s)
```

```
6
```

```
lst = [3,5,8,1]
len(lst)
```

```
4
```

```
d = {"name" : "Raj", "age": 22, "city":"Mumbai"}
len(d)
```

```
3
```

```
1 + 2 #addition
```

```
3
```

```
"a" + "b" #concatenation
```

```
'ab'
```

DPP-Questions

```
class Car:

    def __init__(self, brand):
        self.brand = brand

car1 = Car("BMW")
car2 = car1

car2.brand = "Audi"

print(car1.brand)
```

```
class Student:
    pass

s1 = Student()
s2 = Student()
s3 = s2
```

```
class Movie:

    def __init__(self, name):
        self.name = name

movie = Movie("Avatar")

print(hasattr(movie, "rating"))
```

```
class Student:

    def __init__(self):
        self.age = 20

student = Student()

print(student.age + 5)
```

```
class Animal:

    def sound(self):
        print("Some Sound")

dog = Animal()

dog.sound()
dog.sound()
dog.sound()
```

```
class Laptop:

    def __init__(self, brand, ram):
        self.brand = brand
        self.ram = ram

laptop = Laptop("Dell", 16)

print(laptop.brand, laptop.ram)
```

```
class Book:

    def __init__(self, title):
        self.title = title

book1 = Book("Python")
book2 = Book("SQL")

print(book1.title == book2.title)
```

```
class Student:

    def greet(self):
        return "Welcome"

student = Student()

message = student.greet()

print(message)
```

```
class Bank:

    def __init__(self, balance):
        self.balance = balance

    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount

account = Bank(1000)

account.withdraw(200)
account.withdraw(300)

print(account.balance)
```